

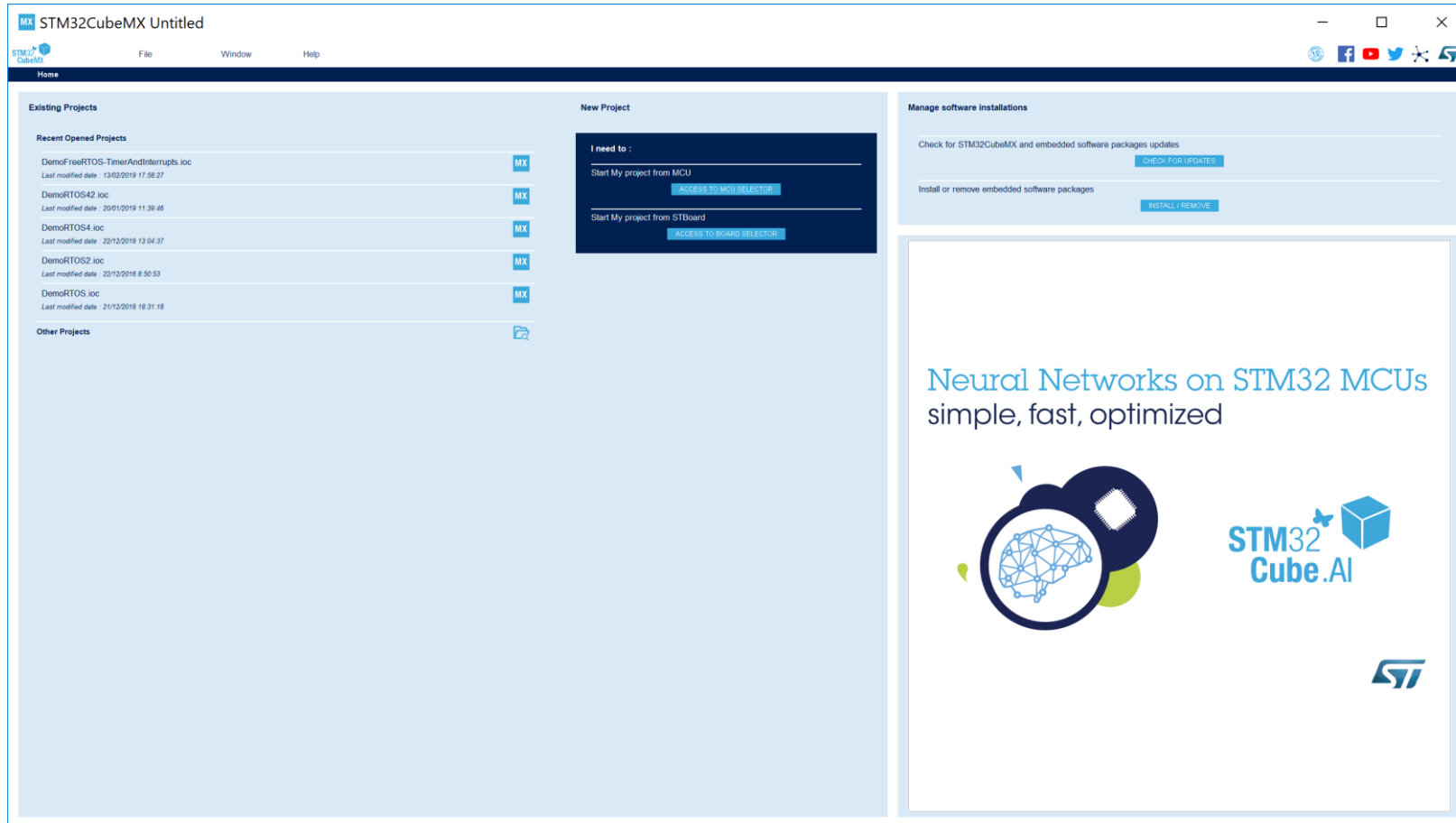
# USCD Embedded C Assignment 2

By

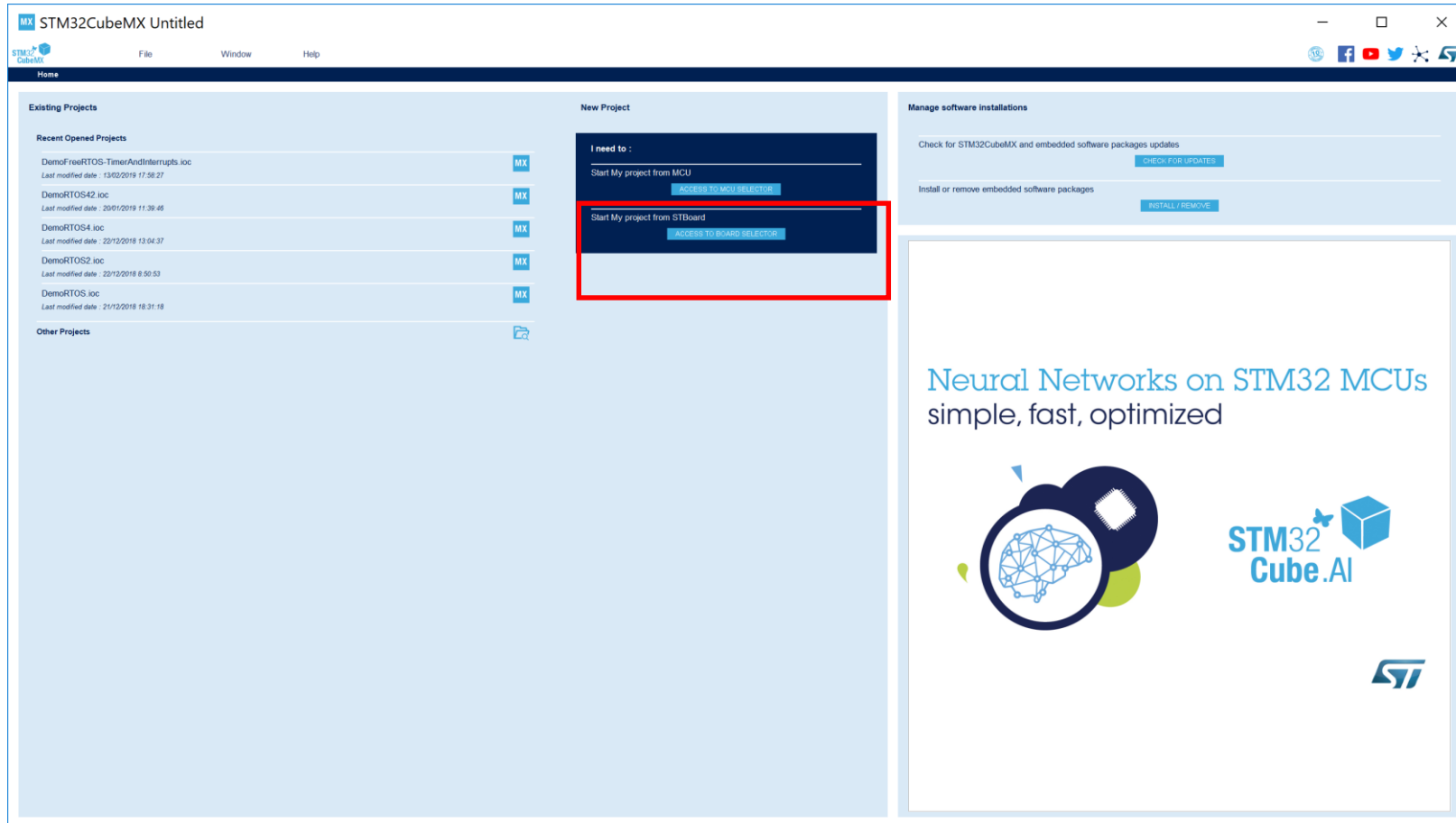
Norman McEntire

[Norman.mcentire@gmail.com](mailto:Norman.mcentire@gmail.com)

# Step 1. Startup STM32CubeMX



# Step 2. Access Board Selector



# Step 3. Enter “B-L475E-IOT01A” Board

**New Project from a Board**

**Board Filter**

Product Search:

Vendor:

Type:

MCU Series:

Other:

Peripheral:

- Accelerometer
- ADC
- Audio Codec
- Audio Line In
- Audio Line Out
- Button
- Camera
- Compass
- Custom Firm. Fw
- Display
- External
- Gyroscope
- LED
- Magnetometer
- Microphone
- On-board Debug
- Other
- Power Source
- Pressure Sensor
- ROM
- RS-232
- RTC
- SmartCard
- Temperature Sensor
- USB

**B-L475E-IOT01A**

**STMicroelectronics B-L475E-IOT01A IOT Discovery Board Support and Examples**

**ACTIVE** Active  
Product is in mass production

Unit Price (USD): \$3.8  
Mounted device: [STM32L475G5](#)

The B-L475E-IOT01A Discovery kit for IoT node allows users to develop applications with direct connection to cloud servers.  
The Discovery kit enables a wide diversity of applications by exploiting the power communication, multi-sensing and ARM Cortex-M4 core-based STM32L4 Series features.  
The support for Arduino Uno V3 and PMOD connectivity provides unlimited expansion capabilities with a large choice of specialized add-on boards.

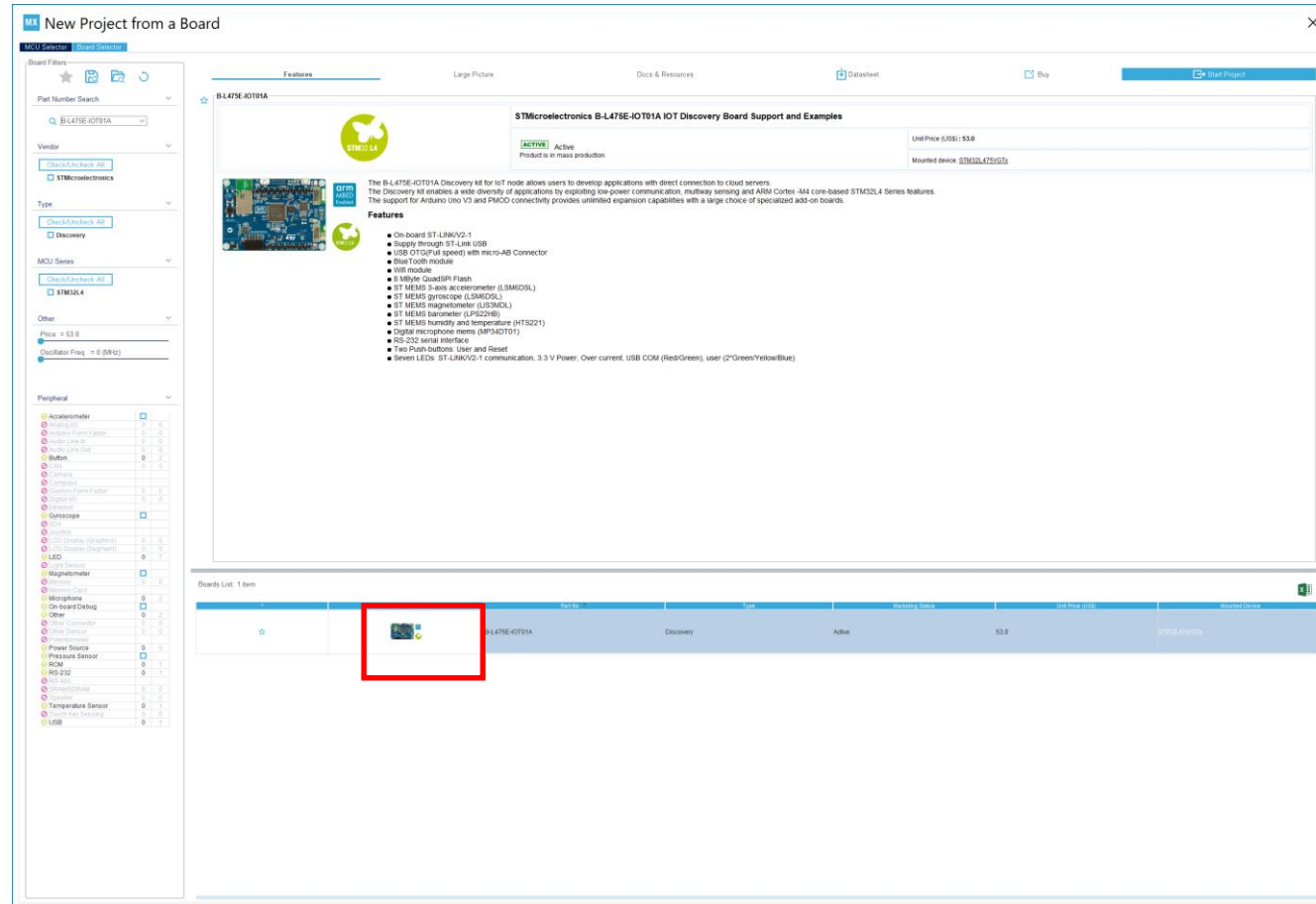
**Features**

- On-board ST-LINKV2-1
- Supply through ST-Link USB
- USB OTG (Full speed) with micro-AB Connector
- Blue Tooth module
- WiFi module
- 8 MByte QuadSPI Flash
- ST MEMS 3-axis accelerometer (LSM2DS1)
- ST MEMS gyroscope (LSM2DS1)
- ST MEMS magnetometer (LSM2DS1)
- ST MEMS barometer (LPS22DH)
- ST MEMS humidity and temperature (HTS221)
- Digital microphone (MP34DT01)
- RS-232 serial interface
- Two Push-buttons: User and Reset
- Seven LEDs: ST-LINKV2-1 communication, 3.3 V Power, Over current, USB COM (Red/Green), user (2\*Green/Yellow/Blue)

**Boards List: 1 item**

| Image | Chipset        | Type      | Status & Link | Unit Price (USD) | Mounted device              |
|-------|----------------|-----------|---------------|------------------|-----------------------------|
|       | B-L475E-IOT01A | Discovery | Active        | \$3.8            | <a href="#">stm32l475g5</a> |

## Step 4. Select Board Photo



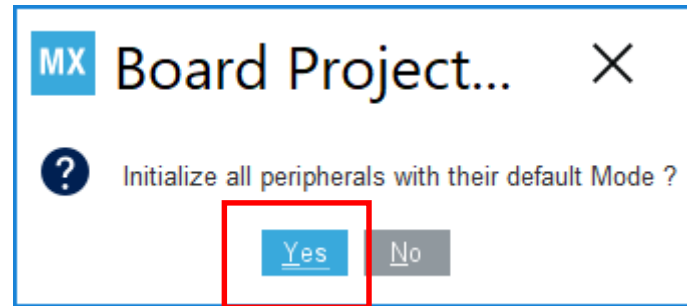
# Step 5. Select “Start Project”

The screenshot shows the 'New Project from a Board' dialog in an IDE. The dialog is divided into several sections:

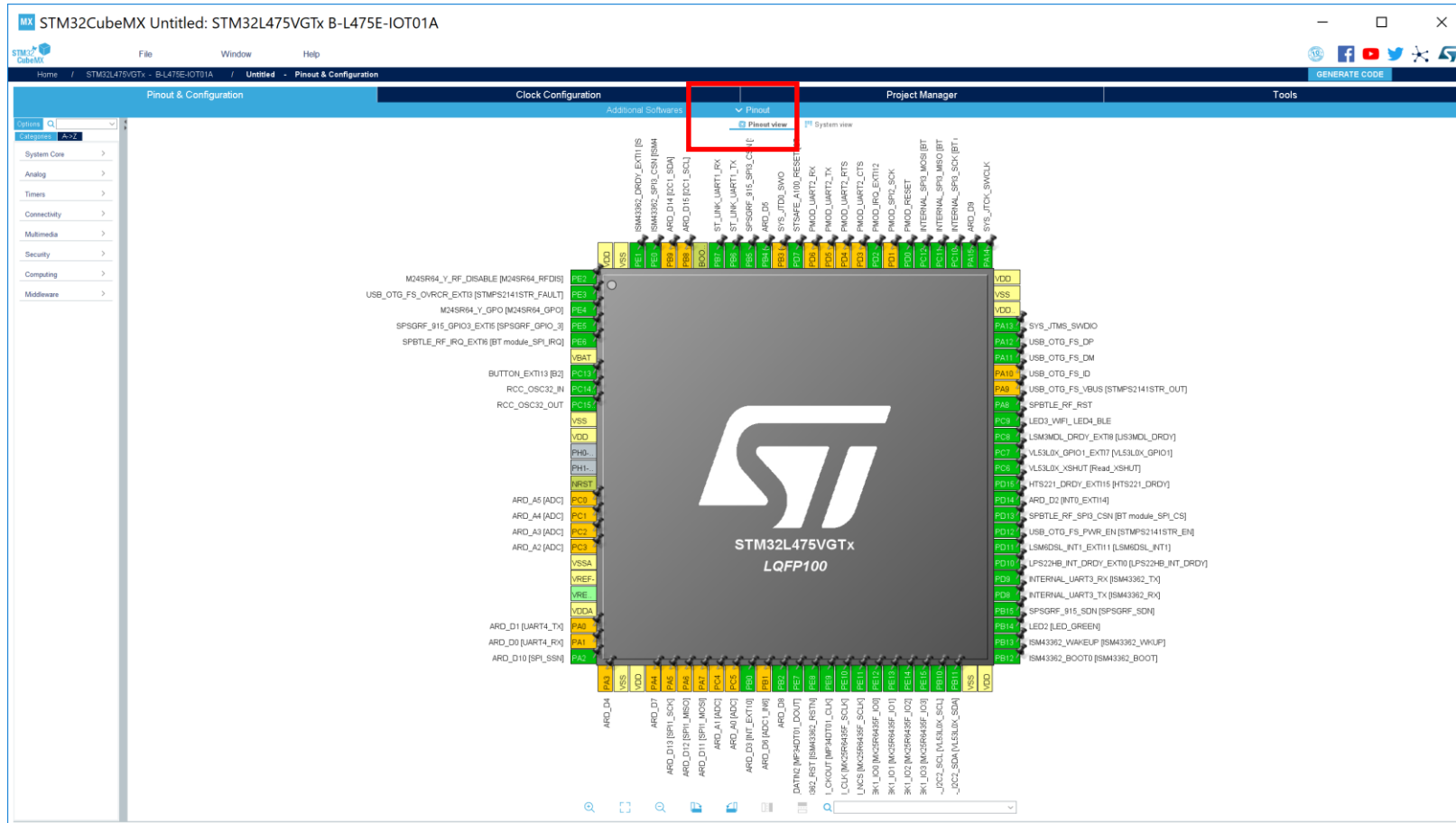
- Board Filters:** A sidebar on the left with filters for Part Number Search, Vendor, Type, MCU Series, Other, and Peripheral. The 'Peripheral' section is expanded, showing a list of components like Accelerometer, Button, Gyroscope, Magnetometer, Microphone, On-board Display, Power Source, Pressure Sensor, ROM, RS-232, RTC, Temperature Sensor, and USB.
- Board Selection:** A central area displaying the selected board, 'B-L475E-IOT01A'. It includes a large picture of the board, a list of features, and a 'Start Project' button highlighted with a red rectangle.
- Boards List:** A table at the bottom showing a list of boards. The first row is highlighted, corresponding to the selected board.

The 'Start Project' button is located in the top right corner of the board selection area, and is highlighted with a red rectangle.

Step 6. Select **YES** (initialize all peripherals with the default mode)

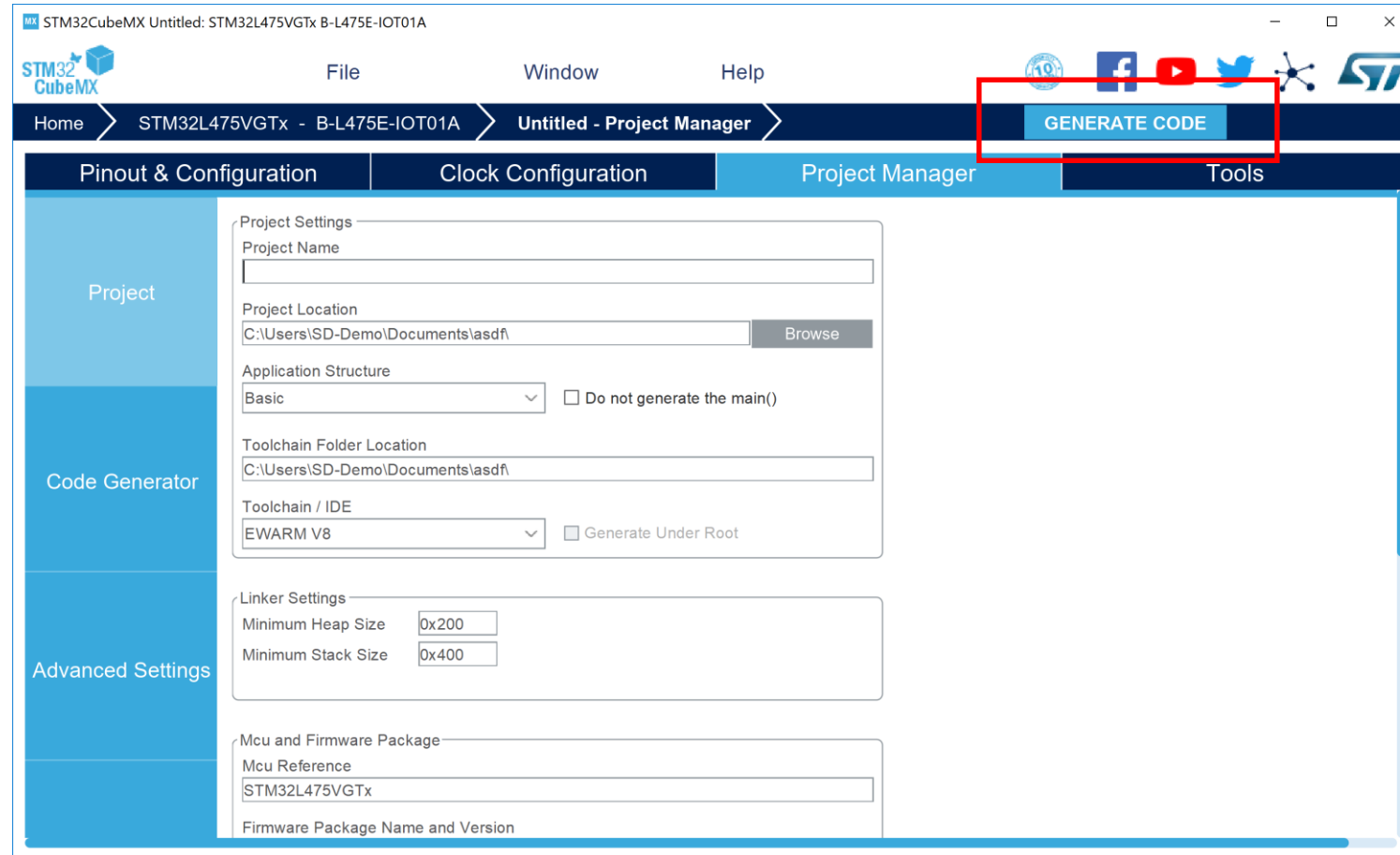


# Step 7. Observe Results (Pinout View)





# Step 8. Select Project Manager Tab



# Step 9. Select Advanced Settings

The screenshot shows the STM32CubeMX software interface. The title bar indicates the project is 'STM32CubeMX Untitled: STM32L475VGTx B-L475E-IOT01A'. The menu bar includes 'File', 'Window', and 'Help'. The breadcrumb navigation shows 'Home > STM32L475VGTx - B-L475E-IOT01A > Untitled - Project Manager'. A 'GENERATE CODE' button is visible in the top right. The main interface has four tabs: 'Pinout & Configuration', 'Clock Configuration', 'Project Manager' (selected), and 'Tools'. On the left, a sidebar contains 'Project' and 'Code Generator' sections. The 'Advanced Settings' option is highlighted with a red box. The main area displays a 'Driver Selector' with a search bar and a list of drivers and their HAL components. Below this is a 'Generated Function Calls' table.

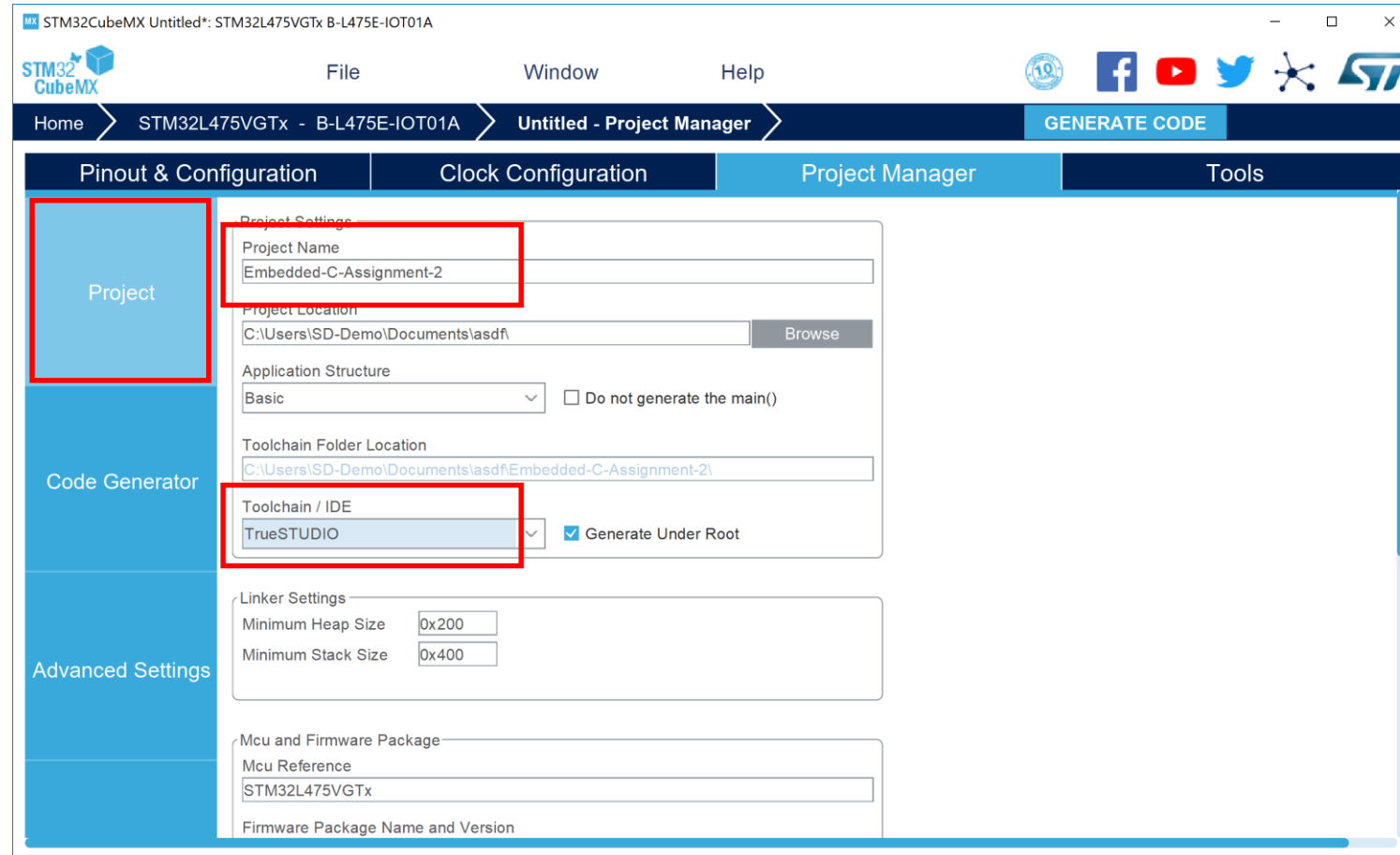
| Rank | Function Name          | IP Instance Name | <input type="checkbox"/> Not Generate Function C... | <input type="checkbox"/> Visibility (Static) |
|------|------------------------|------------------|-----------------------------------------------------|----------------------------------------------|
| 1    | MX_GPIO_Init           | GPIO             | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 2    | SystemClock_Config     | RCC              | <input type="checkbox"/>                            | <input type="checkbox"/>                     |
| 3    | MX_DFSDM1_Init         | DFSDM1           | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 4    | MX_I2C2_Init           | I2C2             | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 5    | MX_QUADSPI_Init        | QUADSPI          | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 6    | MX_SPI3_Init           | SPI3             | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 7    | MX_USART1_UART_Init    | USART1           | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 8    | MX_USART3_UART_Init    | USART3           | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |
| 9    | MX_USB_OTG_FS_PCD_Init | USB_OTG_FS       | <input type="checkbox"/>                            | <input checked="" type="checkbox"/>          |

# Step 10. For Driver Selector, select LL for all that are available with LL option

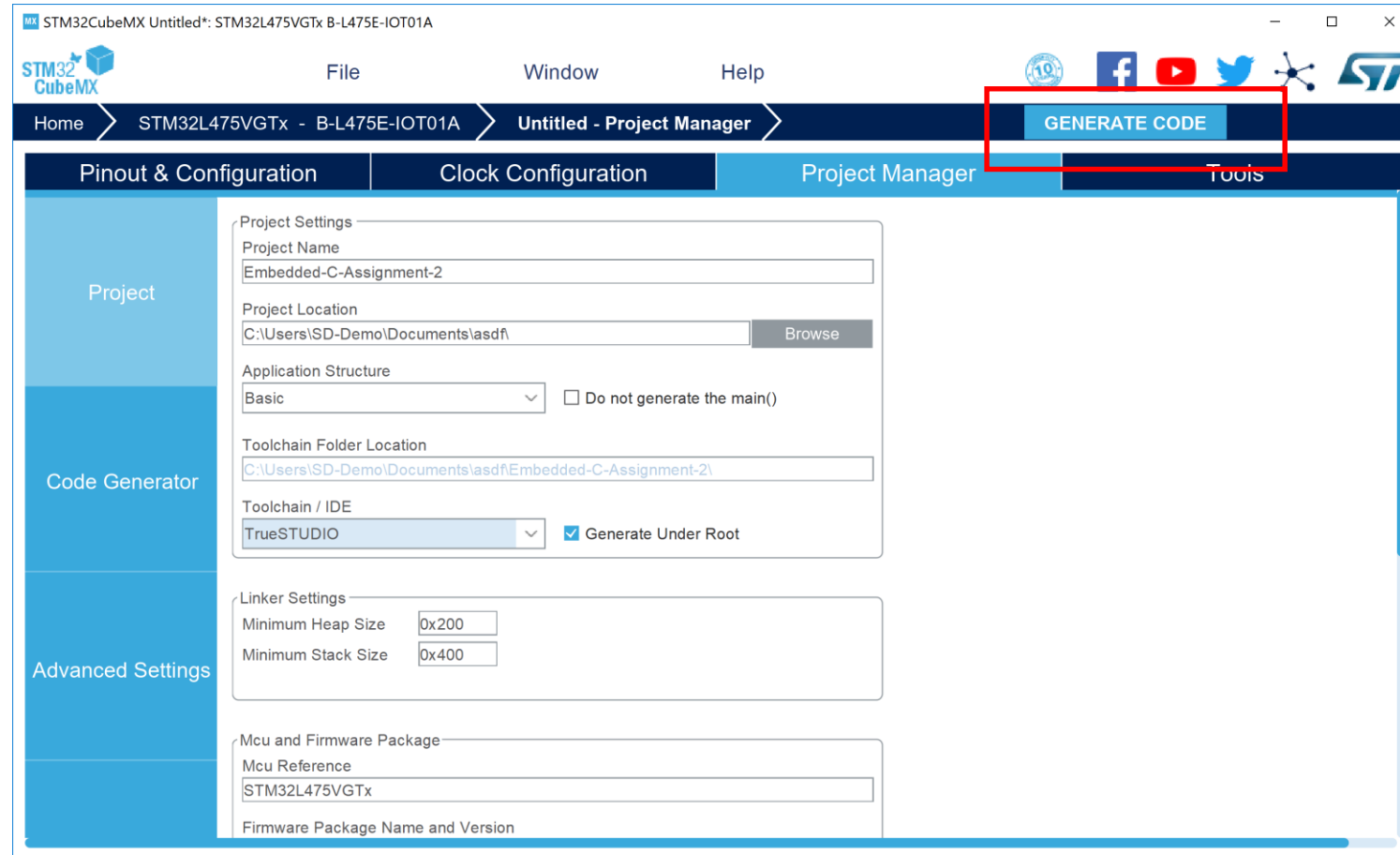
The screenshot shows the STM32CubeMX software interface. The top menu bar includes File, Window, and Help. The main toolbar has buttons for Home, STM32L475VGTx - B-L475E-IOT01A, Untitled - Project Manager, and GENERATE CODE. The left sidebar contains Project, Code Generator, and Advanced Settings. The main area is divided into four tabs: Pinout & Configuration, Clock Configuration, Project Manager, and Tools. The Project Manager tab is active, showing the Driver Selector section. A red box highlights the list of drivers: USART, RCC, DFSDM, USB\_OTG\_FS, I2C, QUADSPI, SPI, and GPIO. The list of drivers is: LL, LL, HAL, HAL, LL, HAL, LL, LL. Below the Driver Selector is the Generated Function Calls section, which contains a table with 9 rows and 5 columns: Rank, Function Name, IP Instance Name, Not Generate Function C..., and Visibility (Static).

| Rank | Function Name          | IP Instance Name | Not Generate Function C... | Visibility (Static)                 |
|------|------------------------|------------------|----------------------------|-------------------------------------|
| 1    | MX_GPIO_Init           | GPIO             | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 2    | SystemClock_Config     | RCC              | <input type="checkbox"/>   | <input type="checkbox"/>            |
| 3    | MX_DFSDM1_Init         | DFSDM1           | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 4    | MX_I2C2_Init           | I2C2             | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 5    | MX_QUADSPI_Init        | QUADSPI          | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 6    | MX_SPI3_Init           | SPI3             | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 7    | MX_USART1_UART_Init    | USART1           | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 8    | MX_USART3_UART_Init    | USART3           | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |
| 9    | MX_USB_OTG_FS_PCD_Init | USB_OTG_FS       | <input type="checkbox"/>   | <input checked="" type="checkbox"/> |

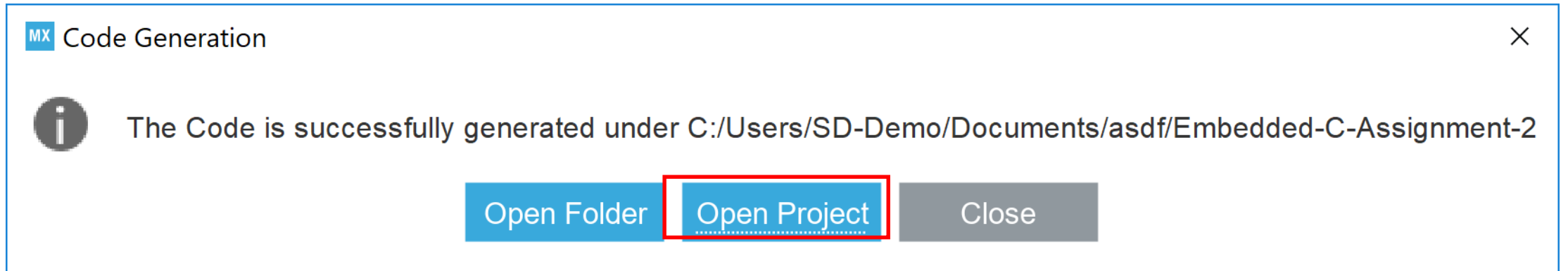
# Step 11. Enter “Embedded-C-Assignment-2” and select TrueStudio as IDE



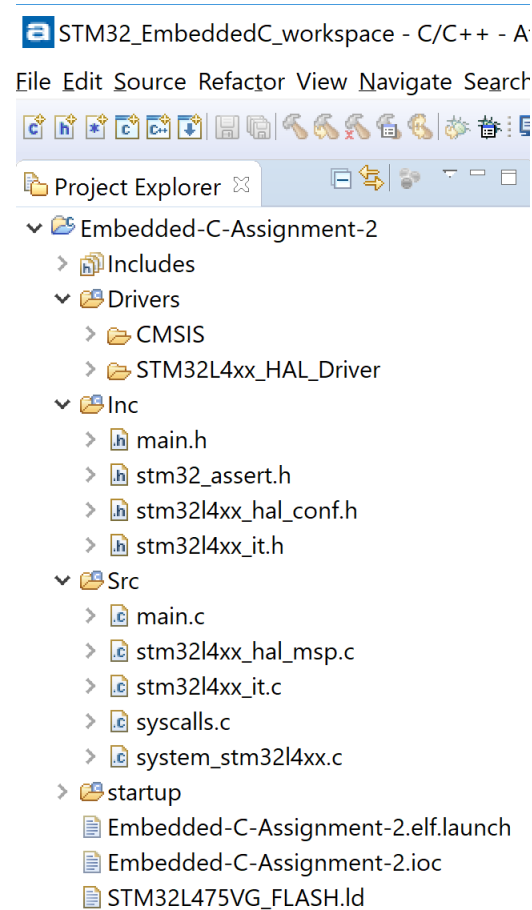
# Step 12. Select “Generate Code”



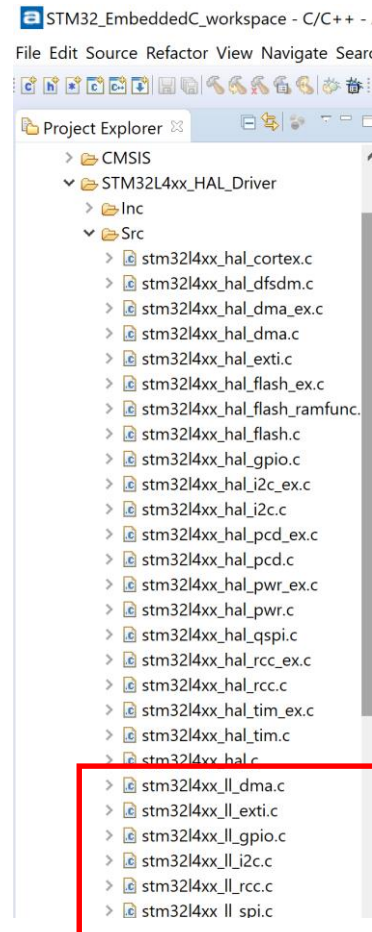
# Step 13. Select “Open Project”



# Step 14. Resulting Project

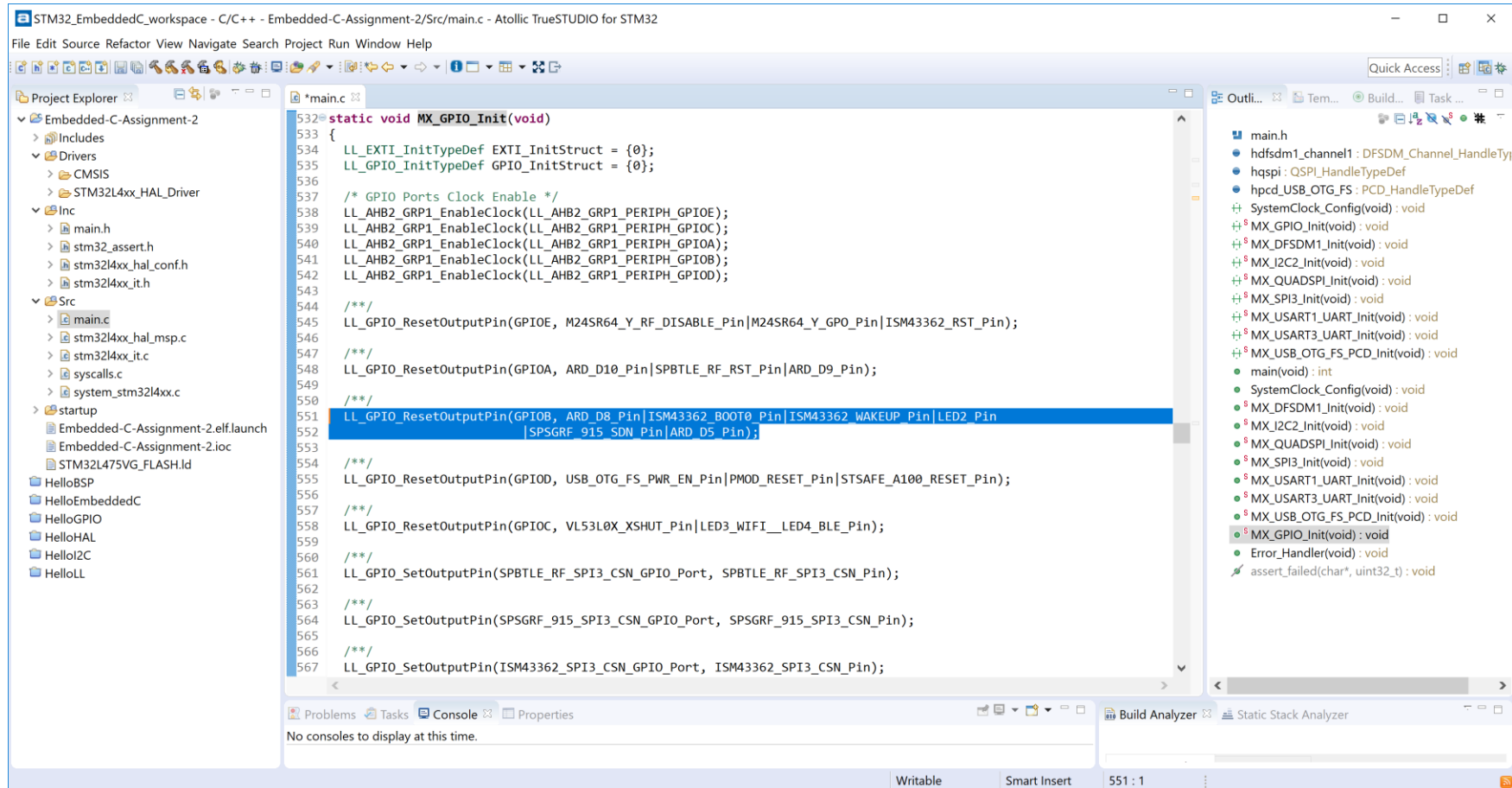


# Step 15. Confirm that LL drivers added to project

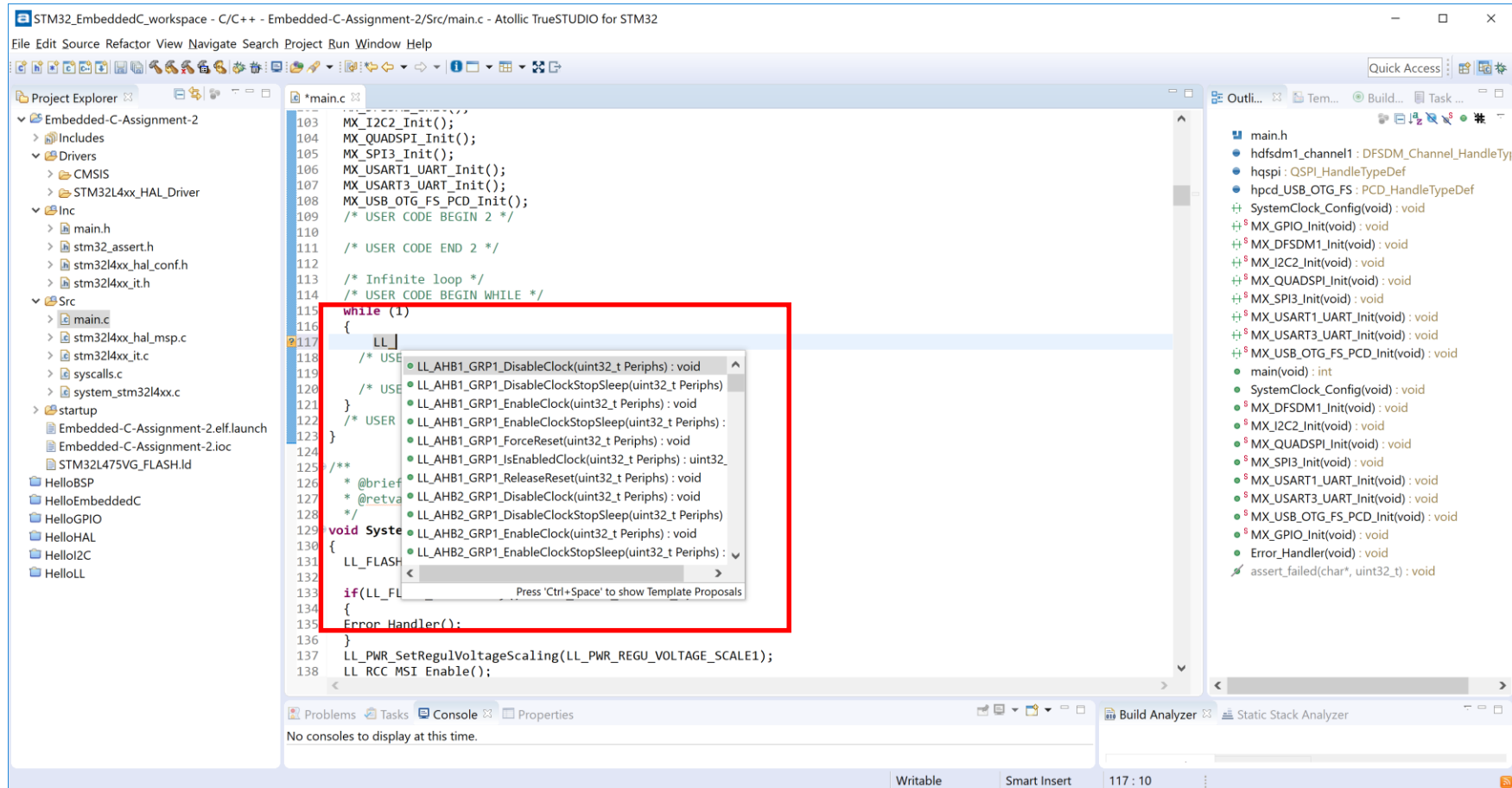




# Step 16. Find GPIO Init code that initializes the LED2 using LL (highlighted in blue below)



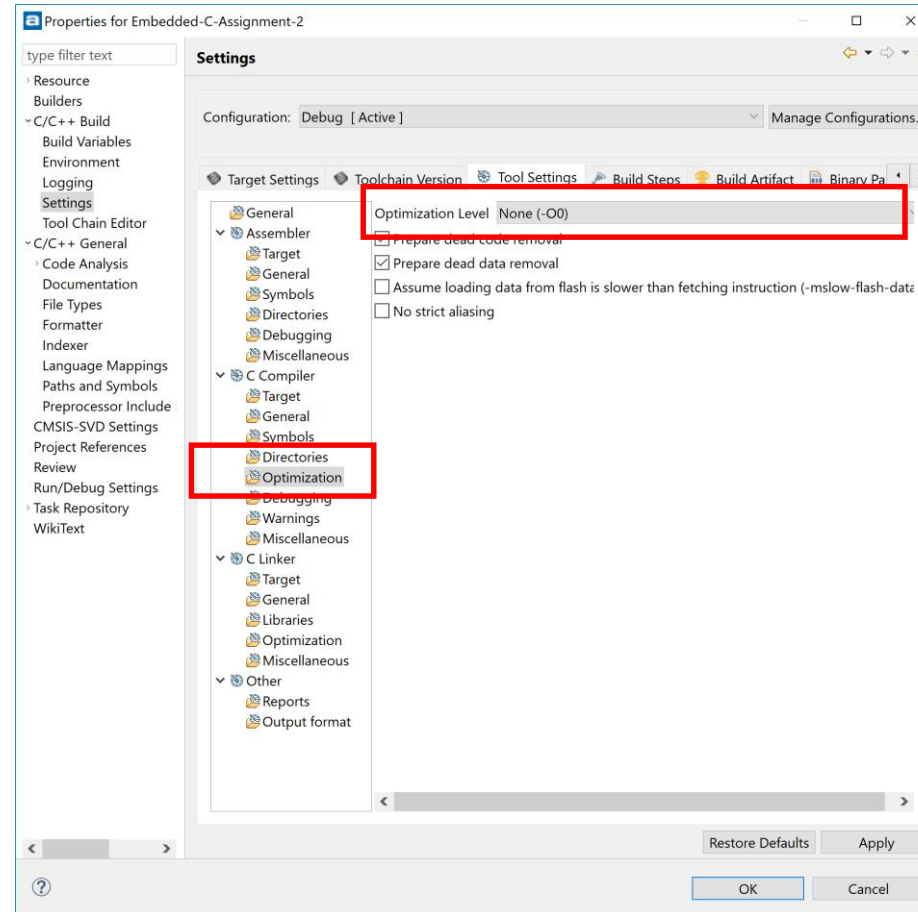
Step 17. In main.c, inside the “while(1)” loop, enter “LL\_” then press Ctrl+SpaceBar to observe LL\_ APIs



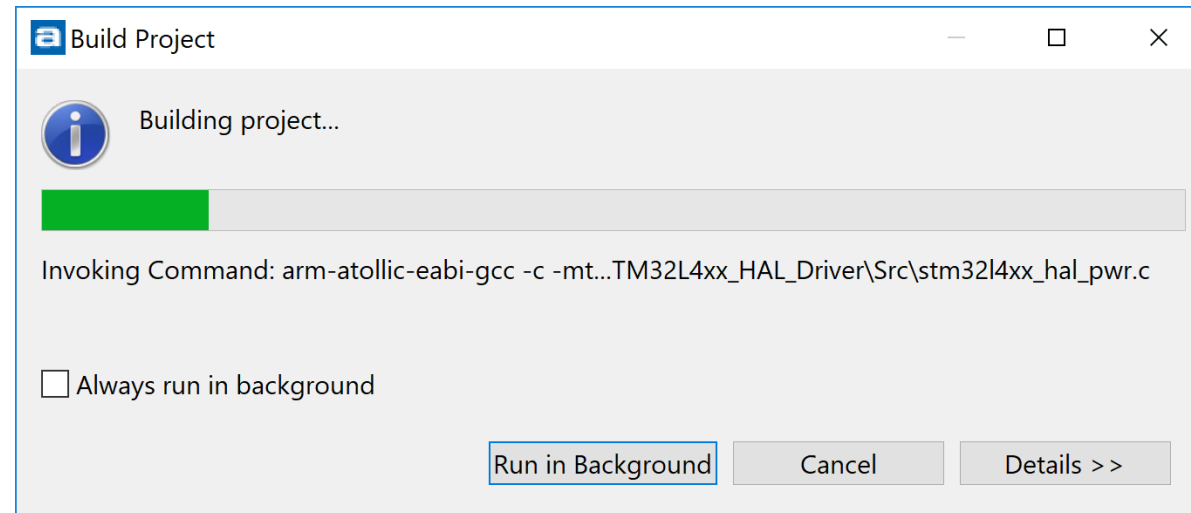
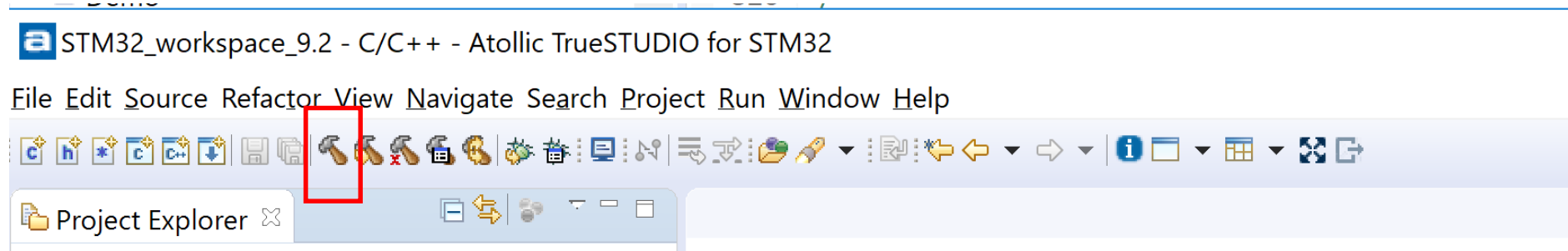
Step 18. In main.c, enter LL related code shown below

```
main.c
110
111 /* USER CODE END 2 */
112
113 /* Infinite loop */
114 /* USER CODE BEGIN WHILE */
115
116 uint32_t flash_size = LL_GetFlashSize();
117 printf("flash_size: %lx\n", flash_size);
118
119 uint32_t uid[3];
120 uid[0] = LL_GetUID_Word0();
121 uid[1] = LL_GetUID_Word1();
122 uid[2] = LL_GetUID_Word2();
123 printf("uid: %lx %lx %lx\n", uid[0], uid[1], uid[2]);
124
125 while (1)
126 {
127     LL_GPIO_TogglePin(GPIOB, LED2_Pin);
128
129     LL_mDelay(1000);
130
131     /* USER CODE END WHILE */
132
133 /* USER CODE BEGIN 3 */
134 }
```

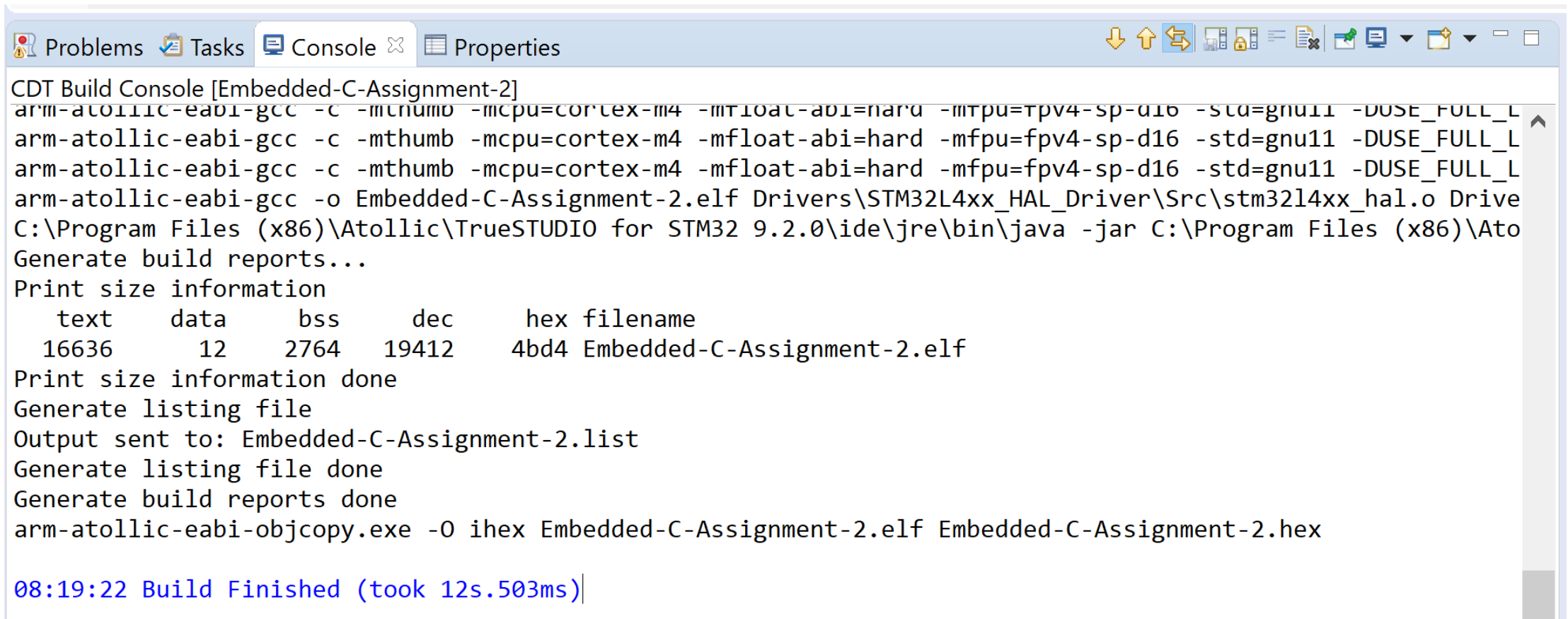
# Step 19. Properties, C/C++ Build, Settings, C Compiler, Optimization, None



# Step 20. Build Project



# Step 21. Results of Build – Part 1



The screenshot shows the CDT Build Console window for 'Embedded-C-Assignment-2'. The console displays the compilation command for 'stm32l4xx\_hal.o', the generation of build reports, size information, and the final build completion message.

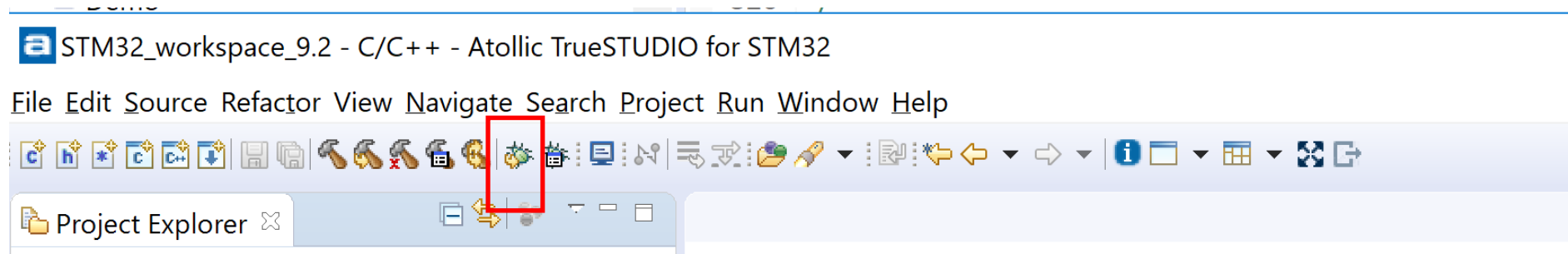
```
CDT Build Console [Embedded-C-Assignment-2]
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -o Embedded-C-Assignment-2.elf Drivers\STM32L4xx_HAL_Driver\Src\stm32l4xx_hal.o Drive
C:\Program Files (x86)\Atollic\TrueSTUDIO for STM32 9.2.0\ide\jre\bin\java -jar C:\Program Files (x86)\Ato
Generate build reports...
Print size information
  text    data    bss     dec     hex filename
 16636     12   2764   19412   4bd4 Embedded-C-Assignment-2.elf
Print size information done
Generate listing file
Output sent to: Embedded-C-Assignment-2.list
Generate listing file done
Generate build reports done
arm-atollic-eabi-objcopy.exe -O ihex Embedded-C-Assignment-2.elf Embedded-C-Assignment-2.hex

08:19:22 Build Finished (took 12s.503ms)
```

# Step 22. Results of Build – Part 2

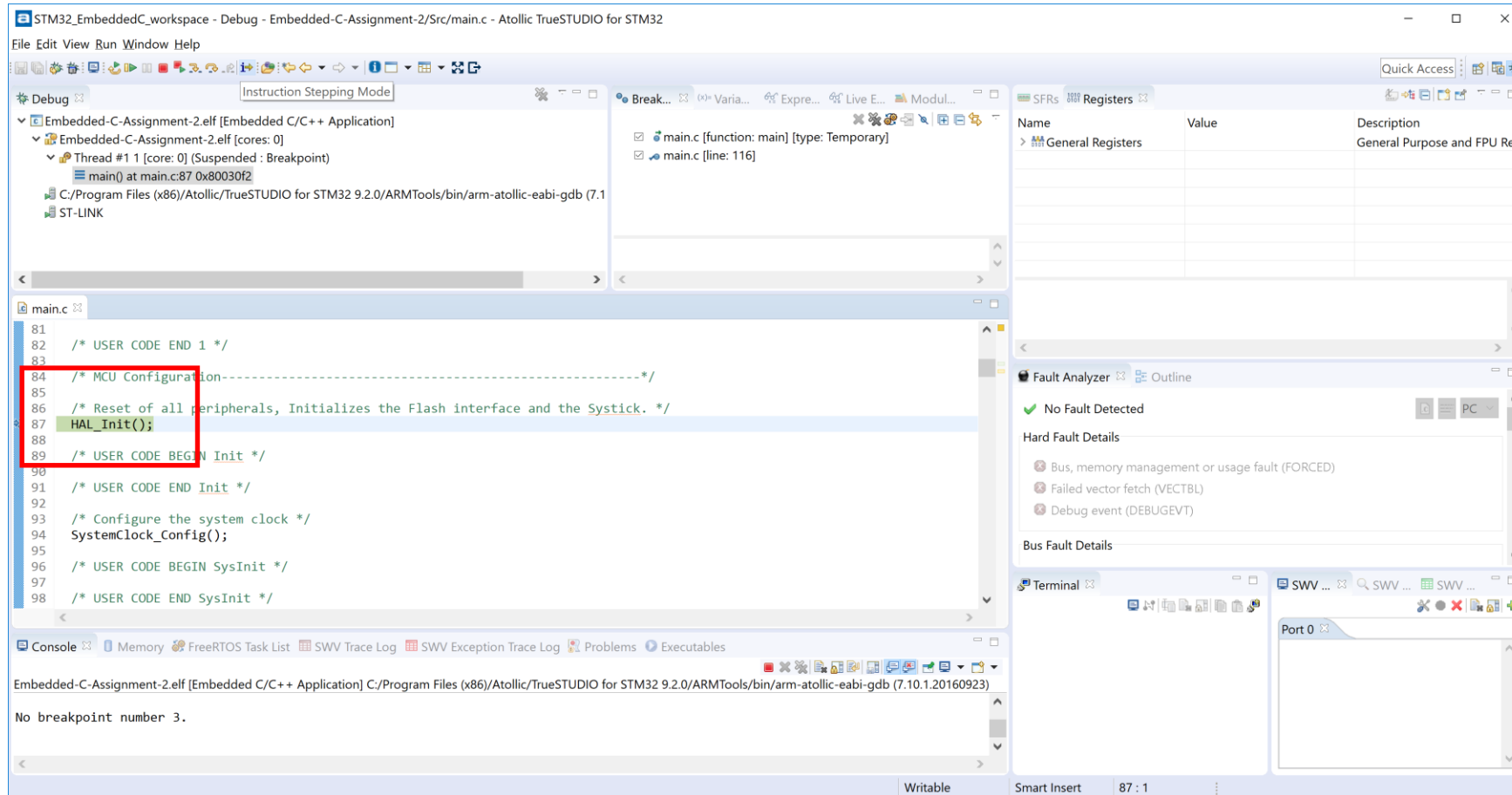
| Memory Regions                                                                          |               | Memory Details |         |            |          |                              |  |
|-----------------------------------------------------------------------------------------|---------------|----------------|---------|------------|----------|------------------------------|--|
| Region                                                                                  | Start address | End address    | Size    | Free       | Used     | Usage (%)                    |  |
|  RAM   | 0x20000000    | 0x20018000     | 96 KB   | 93.18 KB   | 2.82 KB  | <div><div></div></div> 2.93% |  |
|  RAM2  | 0x10000000    | 0x10008000     | 32 KB   | 32 KB      | 0 B      | 0.00%                        |  |
|  FLASH | 0x08000000    | 0x08100000     | 1024 KB | 1003.48 KB | 20.52 KB | <div><div></div></div> 2.00% |  |
|                                                                                         |               |                |         |            |          |                              |  |
|                                                                                         |               |                |         |            |          |                              |  |
|                                                                                         |               |                |         |            |          |                              |  |

# Step 23. Run in Debug

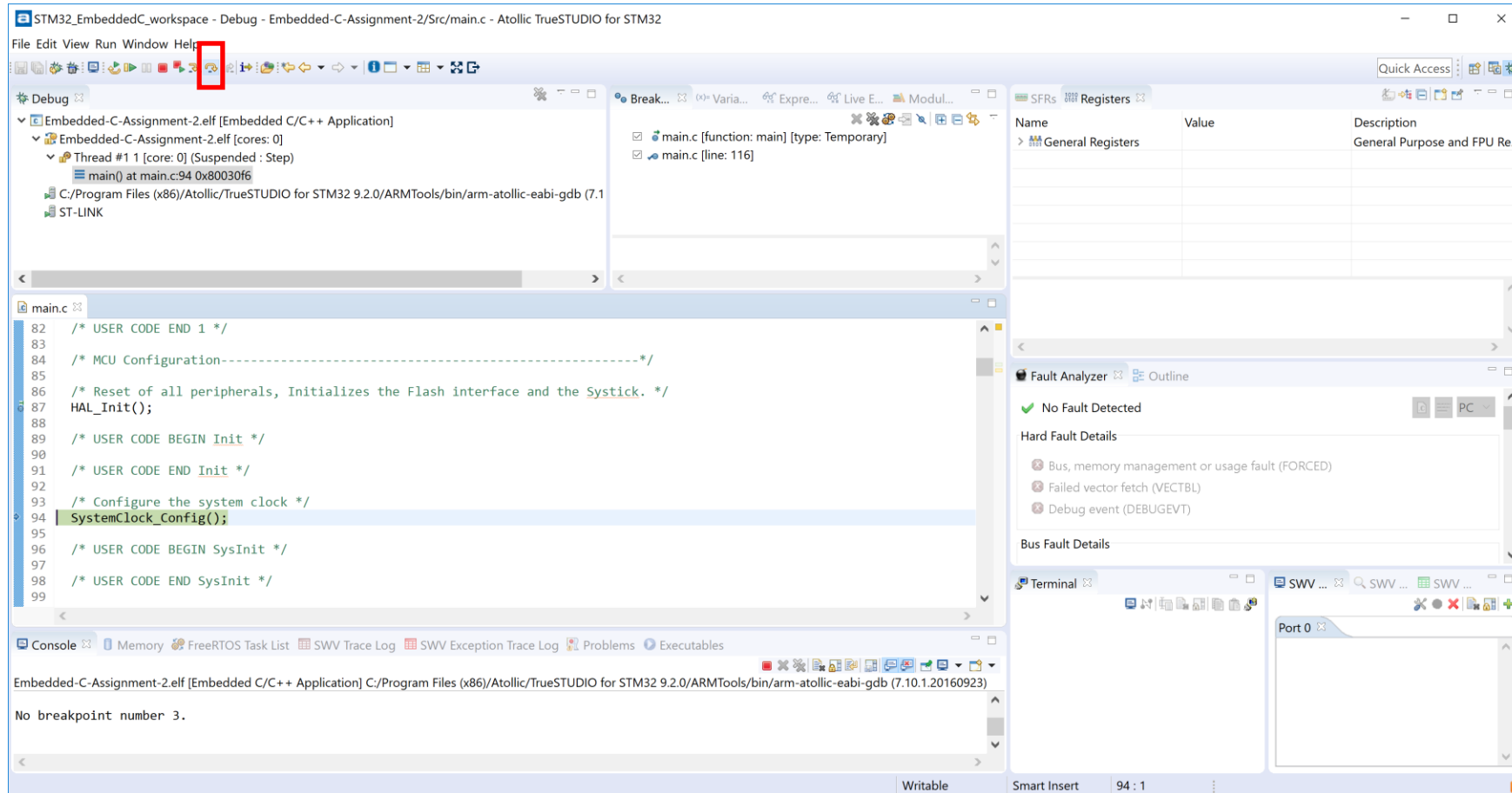




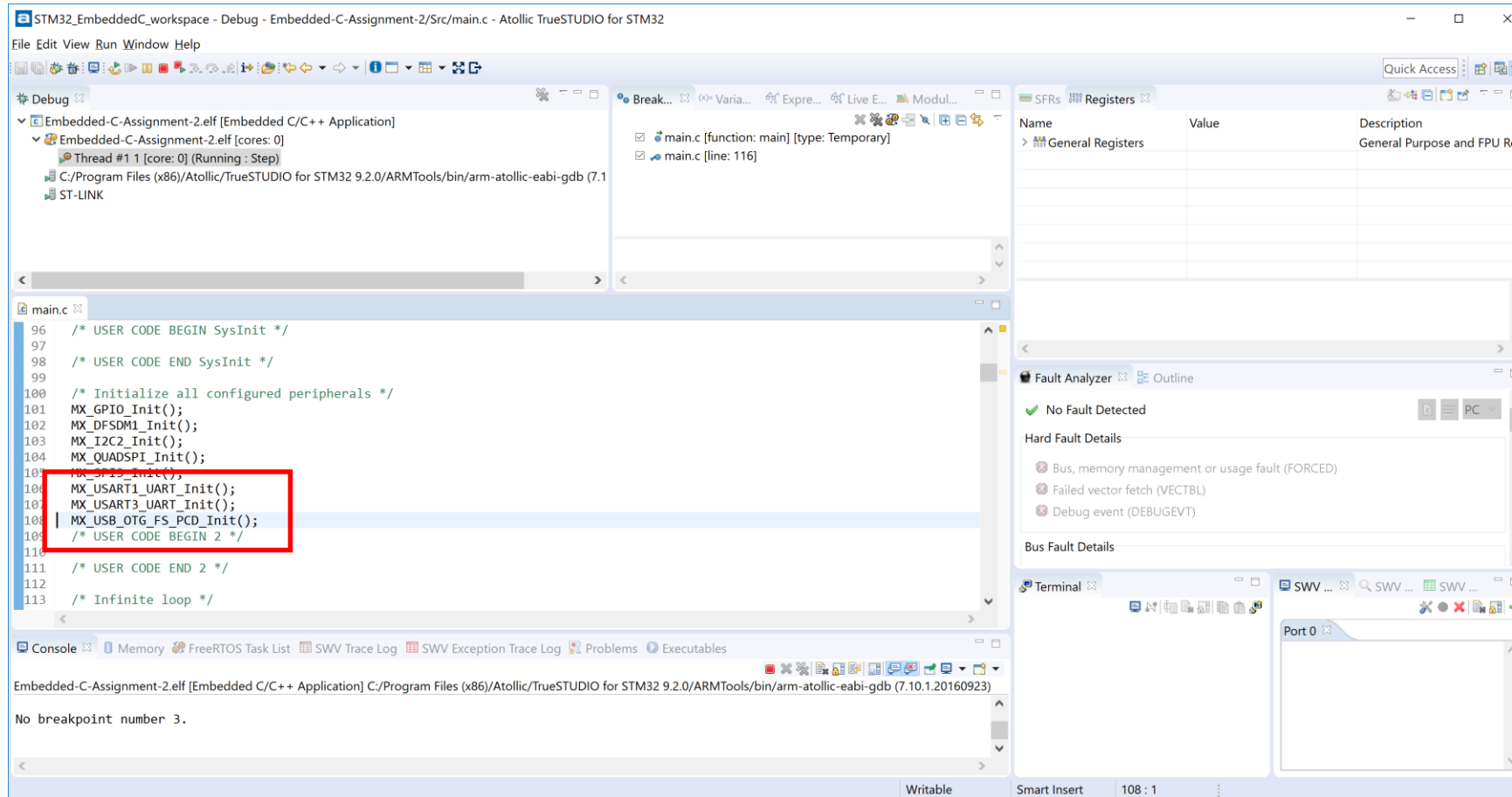
# Step 24. Hit Breakpoint



# Step 25. Click “Step Over”.



Step 26. Click “Step Over”. Repeat as needed.  
Mine “hung” at MX\_USB\_OTG\_FS\_PCD\_Init()



# Step 27. Comment out MX\_USB\_OTG\_FS\_PCD\_Init(), then rebuild

```
*main.c
96  /* USER CODE BEGIN SysInit */
97
98  /* USER CODE END SysInit */
99
100 /* Initialize all configured peripherals */
101 MX_GPIO_Init();
102 MX_DFSDM1_Init();
103 MX_I2C2_Init();
104 MX_QUADSPI_Init();
105 MX_SPI3_Init();
106 MX_USART1_UART_Init();
107 MX_USART3_UART_Init();
108 //NOT USING USB - THIS FUNCTION HUNG MX_USB_OTG_FS_PCD_Init();
109 /* USER CODE BEGIN 2 */
110
111 /* USER CODE END 2 */
112
113 /* Infinite loop */
114 /* USER CODE BEGIN WHILE */
115
116 uint32_t flash_size = LL_GetFlashSize();
117 printf("flash_size: %lx\n", flash_size);
118
119 uint32_t uid[3];
120 uid[0] = LL_GetUID_Word0();
```

# Step 28. Build, Run, Use Debugger to set through code. Notice LED toggles on/off

